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Machine Learning
Tutorial / Recitation - I

1) Understand & Explain the Data

- Mean / Avg
- Varience / Standard deviation

Sunlight hours (X_1)	water litres (X_2)	Bias (X_3)	Target (good growth = 1, Poor = 0)
3	1	1	0
8	4	1	1
5	2	1	0
10	5	1	1
7	3	1	1
$\frac{33}{5} = 6.6$	$\frac{15}{5} = 3$		

Mean gives the central tendency.

Mean of $X_1 \Rightarrow \mu_{X_1} = 6.6$

Mean of $X_2 \Rightarrow \mu_{X_2} = 3$

Varience $\sigma^2 = \frac{1}{N} \sum_{i} (\underset{\text{Input}}{X_i} - \underset{\text{Mean}}{\mu})^2$

$\sigma^2_{X_1} = (X_{11} - \mu_{X_1})^2 + (X_{12} - \mu_{X_1})^2 + \ldots$

$= \frac{(3-6.6)^2 + (8-6.6)^2 + (5-6.6)^2 + (10-6.6)^2 + (7-6.6)^2}{5}$

$= \frac{12.96 + 1.96 + 2.56 + 11.56 + 0.16}{5}$

$\frac{29.2}{5} = 5.84$

$\sigma^2_{X_2} = \frac{(1-3)^2 + (4-3)^2 + (2-3)^2 + (5-3)^2 + (3-3)^2}{5}$

$= 2.$

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Varience of $\sigma^2_{x_1} = 5.84$ Std. Dev $= \sqrt{\sigma_{x_1}^2} = \sigma_{x_1}$
$= 2.42$

Varience of $\sigma^2_{x_2} = 2$ Std. Dev $= \sqrt{\sigma^2_{x_2}} = \sigma_{x_2}$
$= 1.414$

$\sigma^2_{x_1} > \sigma^2_{x_2}$
$\Downarrow$
It has more
for the target.

ML Model $\Rightarrow$ Perceptron Model
$\Downarrow$
Simple nueral model
$\downarrow$
Simple decision making model
that can answer Yes/no.

Perceptron

1) Inputs: x_1 & x_2
2) weights associated with inputs: w_1 & w_2
3) Bias $\rightarrow$ Model flexibility (b)

4) weighted sum $z = w_1x_1 + w_2x_2 + b$
$= \sum_i w_i x_i + b$

5) Activation function: $\hat{y} = \begin{cases} 1, & \text{if } z \geq 0 \\ 0, & \text{if } z < 0 \end{cases}$
(Step function)

6) Updated weight:
$w_i^{new} = w_i^{old} + \eta (y - \hat{y}) x_i$, $y \rightarrow$ actual target
$\hat{y} \rightarrow$ Predicted target
$\eta \rightarrow$ learning rate.

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Prediction

1) Initial

Let, $w_1 = 0$ Inputs: $x_1 = 3$, $x_2 = 1$, $b = 1$ (bias)

$w_2 = 0$ $z = 3 = 1$ $\hat{y} = 1$

$\eta = 1$ actual $y = 0$.

Prediction ✗

$w_1^{new} = 0 + 1(0-1) \cdot 3 = -3$

$w_2^{new} = 0 + 1(0-1) \cdot 1 = -1$

now, $w_1 = -3$ $\eta = 1$

$w_2 = -1$ $b = 1$

Inputs $x_1 = 8$, $x_2 = 4$, $b = 1$

$z = w_1 x_1 + w_2 x_2 + b$

$= -24 - 4 + 1$

$= -27$ $z < 0 \Rightarrow \hat{y} = 0$ ✗ Prediction wrong

$w_1^{new} = -3 + 1(1-0) \cdot 8 = 5$

$w_2^{new} = -1 + 1(1-0) \cdot 4 = 3$

Input: $x_1 = 5$ $x_2 = 2$, $b = 1$

$z = 32$. $\hat{y} = 1$ but $y = 0$. Prediction ✗.